

Group 1: Reinforcement Learning in a High-Variance, Combinatorial World

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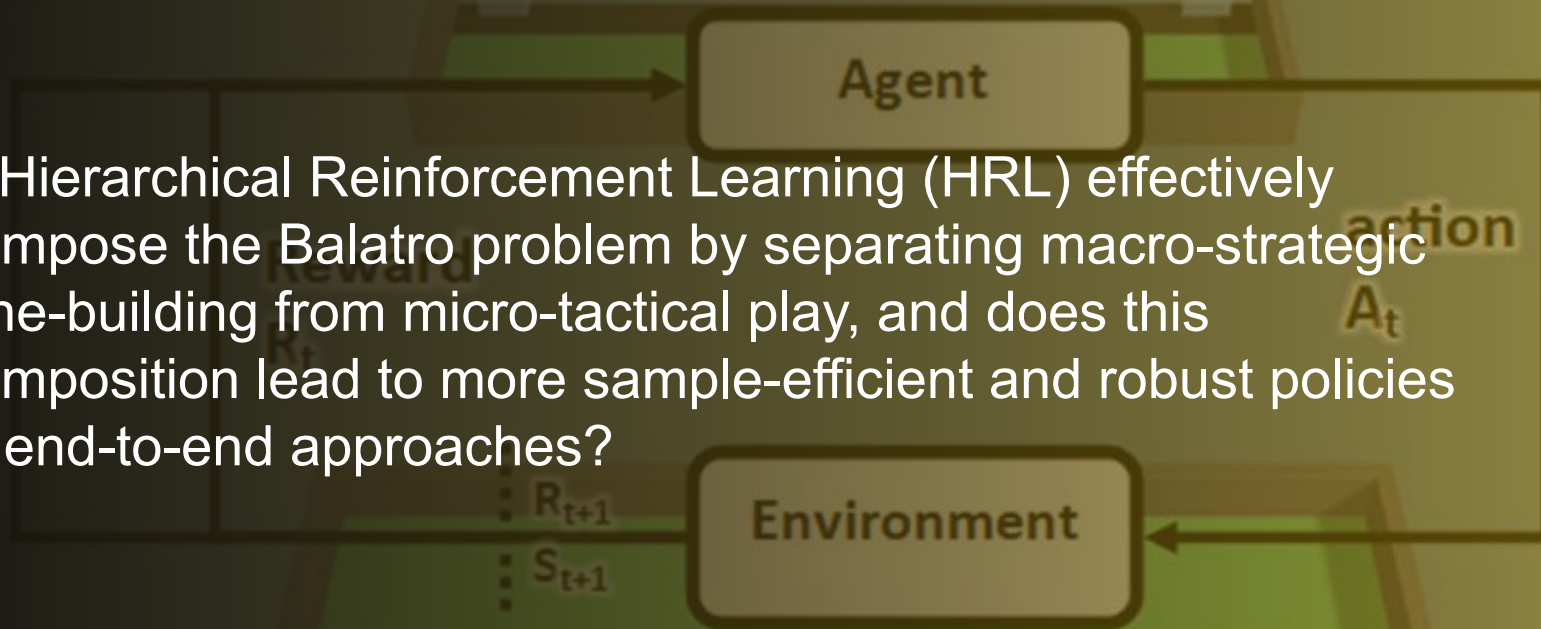
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Research Question

Can Hierarchical Reinforcement Learning (HRL) effectively decompose the Balatro problem by separating macro-strategic engine-building from micro-tactical play, and does this decomposition lead to more sample-efficient and robust policies than end-to-end approaches?



Why Balatro?

Simple...

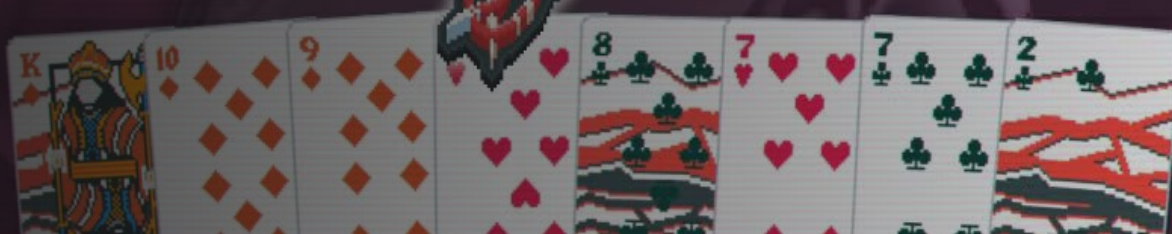
- Main Gameplay is playing poker hands to beat the levels score
- Higher levels -> higher score to pass
- Limited number of Discards and Hands



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BALATRO



Why Balatro?

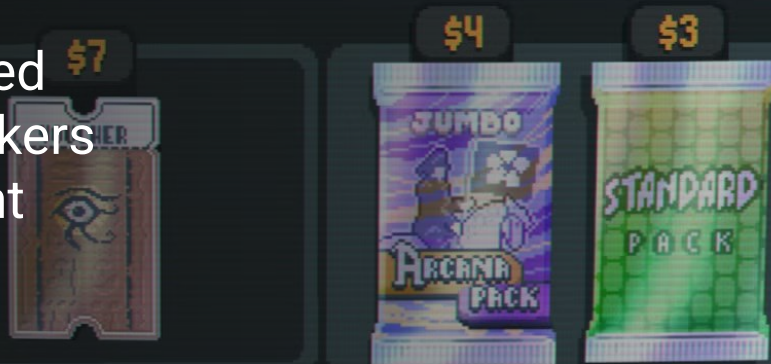
...yet deep

- Has long term strategy and resource management aspects
- Can buy new cards to add to the deck
- Can change how much a hand is worth
- Modify how cards are scored
- Add synergies by adding jokers
- And stochastic environment



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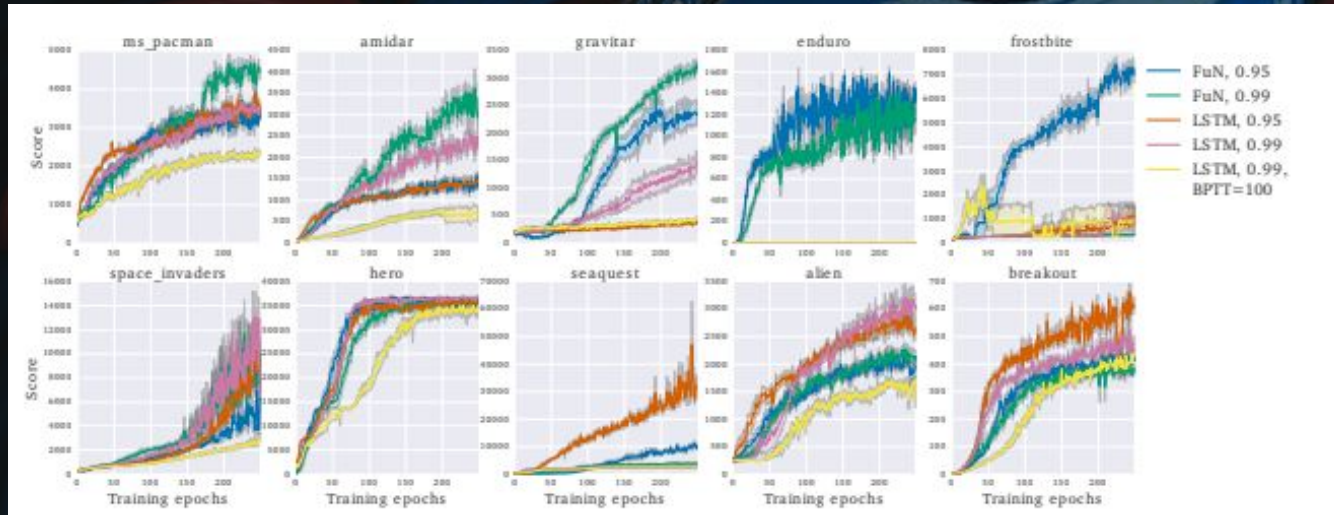
Other Reinforcement Learning games

Feature	Go	No-Limit Texas Hold'em	Slay the Spire	Hearthstone	Balatro
Game Type	Abstract Strategy Board Game	Competitive Card Game	Single-Player Roguelike Deck-BUILDER	Competitive Collectible Card Game	Single-Player Roguelike Deck-BUILDER
Information Structure	Perfect	Imperfect (Opponent's cards)	Imperfect (Deck order, event outcomes)	Imperfect (Opponent's hand/deck)	Imperfect (Deck order, shop offerings)
Key Source of Stochasticity	None (Deterministic)	Card Deal	Card Draw, Relic/Potion Rewards, Map Generation	Card Draw, Random Card Effects	Card Draw, Shop Offerings, Boss/Tag Selection
Primary AI Challenge	Vast search space, long-term positional evaluation	Opponent modeling, bluffing, game theory optimal play	Pathing, risk management, building synergistic decks against varied enemies	Two-phase strategy (deck-building and play), handling complex card semantics	Long-horizon credit assignment in a high-variance, synergistic, combinatorial action space; balancing engine-building with tactical adaptation

Paper 1: FeUdal Networks for Hierarchical Reinforcement Learning

Vezhnevets, A. S., Osindero, S., Schaul, T., Heess, N., Jaderberg, M., Silver, D., & Kavukcuoglu, K.

(2017). *FeUdal Networks for Hierarchical Reinforcement Learning*. Proceedings of the 34th International Conference on Machine Learning (ICML).



Core Idea: This paper introduces FeUdal Networks (FuN), an HRL architecture with a **Manager** and a **Worker**. The Manager operates on a lower temporal resolution, setting abstract goals in a learned latent space. The Worker operates at every timestep, receiving these goals and learning to generate actions to satisfy them. The Worker is intrinsically motivated to follow the Manager's direction.

Paper 2: Hierarchical Reinforcement Learning for Real-Time Strategy Games.

Niel, T., & Krebbers, A. (2018). *Hierarchical Reinforcement Learning for Real-Time Strategy Games*. Proceedings of the 10th International Conference on Agents and Artificial Intelligence (ICAART).

Table 3: Mean performance after 26 epochs (games).

Opponent	Method	Reward Application	Win-rate	Tie-rate	Loss-rate	Win:loss
Classic	Q-learning	Individual	19.5%	75.0%	5.5%	7:2
Classic	Q-learning	Shared	13.2%	72.3%	14.4%	9:10
Classic	Monte Carlo method	Individual	18.0%	67.0%	15.0%	6:5
Classic	Monte Carlo method	Shared	21.9%	48.0%	30.0%	7:10
Random	Q-learning	Individual	28.6%	61.5%	10.0%	3:1
Random	Q-learning	Shared	23.5%	57.8%	18.8%	5:4
Random	Monte Carlo method	Individual	28.8%	45.1%	26.2%	11:10
Random	Monte Carlo method	Shared	32.0%	26.2%	41.9%	3:4

Core Idea: This paper applies HRL to a Real-Time Strategy (RTS) game. It uses a high-level RL agent to select among a set of pre-defined, scripted sub-policies (implemented as Finite State Machines - FSMs) like "build base" or "attack enemy." The lower level is not a learned policy but a fixed, programmatic one.

Methodology: First Steps

- Modify besteon/balatrobot to record game state's and player actions into JSON's
- Modify besteon/balatrobot to take in player actions through the API
 - Collecting game state and actions taken during play time
 - ~100 - ~400 gameplay states recorded per session

Methodology: Data

- Collecting game state and actions taken during play time
- ~100 - ~400 gameplay states recorded per session
- Includes cards played, score, shop actions, card rearrangements, and other metadata that might be useful

```

    "state": "DRAW_TO_HAND",
    "jokers": [
      {
        "label": "The Duo"
      }
    ],
    "interest_cap": 35
  },
  {
    "received_timestamp": 172523473.027877,
    "context_type": "shop_session",
    "bankrupt_at": 0,
    "current_state": "state",
    "state": 2,
  },
  {
    "ondeck": "2ma5",
    "name": "name",
    "chips": 0,
    "current": "current",
    "dinds": {
      "ante": {
        "score": 324,
        "goal": 0,
        "hand_scores": {
          "shop_id": "172523473-3408-21-1-172523473",
          "dooff": 4,
          "initiation": 0,
          "hand": [
            "deckback",
            "game_time": 32,
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Methodology: Human Data

- Format Data for machine learning
- Play some games of Balatro and collect player data
- Perform Exploratory Data Analysis to see common human patterns

```
states.sort_values("timestamp")
```

	level	played	#	timestamp	deckback	consumables	state_name	#	discount_percent
0	(Flush House: {'level': 1, 'played': 0, 'total_score': 0})	12	1755219723	0	UNKNOWN				
1	(Flush House: {'level': 1, 'played': 0, 'total_score': 0})	0	1755219723	0	BLIND_SELECT				
2	(Flush House: {'level': 1, 'played': 0, 'total_score': 0})	0	1755219723	0	DRAW_TO_HAND				
3	(Flush House: {'level': 1, 'played': 0, 'total_score': 0})	0	1755219723	0	SELECTING_HAND				
4	(Flush House: {'level': 1, 'played': 0, 'total_score': 0})	0	1755219733	0	HAND_PLAYED				
5	(Flush House: {'level': 1, 'played': 0, 'total_score': 0})	1	1755219736	0	NEW_ROUND				
6	(Flush House: {'level': 1, 'played': 0, 'total_score': 0})	1	1755219749	0	MINI_GAME				
7	(Flush House: {'level': 1, 'played': 0, 'total_score': 0})	1	1755219758	0	SHUFFLE				
8	(Flush House: {'level': 1, 'played': 0, 'total_score': 0})	1	1755219769	0	BUFFOON_PACK				
9	(Flush House: {'level': 1, 'played': 0, 'total_score': 0})	1	1755219769	0	SHOP				

10 rows x 10 cols | 10 | per page

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common human patterns

	id	timestamp	card_info	session_id	action	params		
0	BLIND_SELECT	1755219722_3625_B1-0_1755219722	5	1755219722_3625_B1-0_1755219722	1755219727	0	SELECT_BLIND	0
1	HAND_PLAYED	1755219722_3625_B1-1_1755219722	11	1755219722_3625_B1-1_1755219722	1755219733	('hand_played': {'level': 1, 'total_score': 0})	PLAY_HAND	[3, 4, 5, 7, 8]
2	BUFFOON_PACK	1755219722_3625_S1-1_1755219758	36	Missing value	1755219758	('set': 'Booster', 'key': 'p_buffoon_nor')	USE_BOOSTER_PACK	0
3	BUFFOON_PACK	1755219722_3625_S1-1_1755219758	47	Missing value	1755219769	('set': 'Joker', 'key': 'j_blue_joker', 'hand': 0)	ADD_JOKER	0
4	SPECTRAL_PACK	1755219722_3625_S1-1_1755219772	50	Missing value	1755219772	('set': 'Booster', 'key': 'p_spectral_jum')	USE_BOOSTER_PACK	0
5	SPECTRAL_PACK	1755219722_3625_S1-1_1755219772	61	Missing value	1755219783	('highlighted_hand_positions': [], 'set': 'Booster')	USE_CONSUMABLE	0
6	SHOP	1755219722_3625_S1-1_1755219785	67	Missing value	1755219789	0	END_SHOP	0
7	BLIND_SELECT	1755219722_3625_B1-1_1755219791	69	1755219722_3625_B1-1_1755219791	1755219791	0	SELECT_BLIND	0
8	HAND_PLAYED	1755219722_3625_B1-2_1755219791	74	1755219722_3625_B1-2_1755219791	1755219796	('hand_played': {'level': 1, 'total_score': 0})	PLAY_HAND	[1]
9	DRAW_TO_HAND	1755219722_3625_B1-2_1755219791	82	1755219722_3625_B1-2_1755219791	1755219804	('discarded_cards': [{'suit': 'Hearts', 'rank': 2}])	DISCARD_HAND	[1, 2, 3]

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Methodology: Model Architecture

Manager

Learns and continually updates a high-level, abstract strategy based on the jokers, deck, consumables, money etc. This strategy is sent to the player as a goal (g_t)

$$s_t = f_{Mspace}(x_t)$$

Convert game info to managers state space (s_t) with a fully connected NN layer and ReLU

$$h_t^M, g_t^M = f_{Mrnn}(s_t, h_{t-1}^M)$$

Use a dilated LSTM to get the managers hidden RNN state (h_t^M) and a raw goal vector (g_t^M)

$$g_t = \frac{g_t^M}{\|g_t^M\|}$$

Normalize the goal vector since we only care about direction in latent space

Player

The player will learn the best action (a_t) (hands to play, items to buy, blinds to skip etc.) based on the game state (x_t) its own state (h_t^W) and the managers goal (g_t)

$$h_t^W, U_t = f_{Wrnn}(x_t, h_{t-1}^W)$$

Use a standard LSTM to get the players hidden RNN state (h_t^W) and an action embedding matrix where each row is an action

$$a_t = \text{softmax}(U_t, w_t)$$

Action for the player to take

Convert the last C manager goals into a form that the player can use

$$w_t = \phi \left(\sum_{i=t-C}^t g_i \right)$$

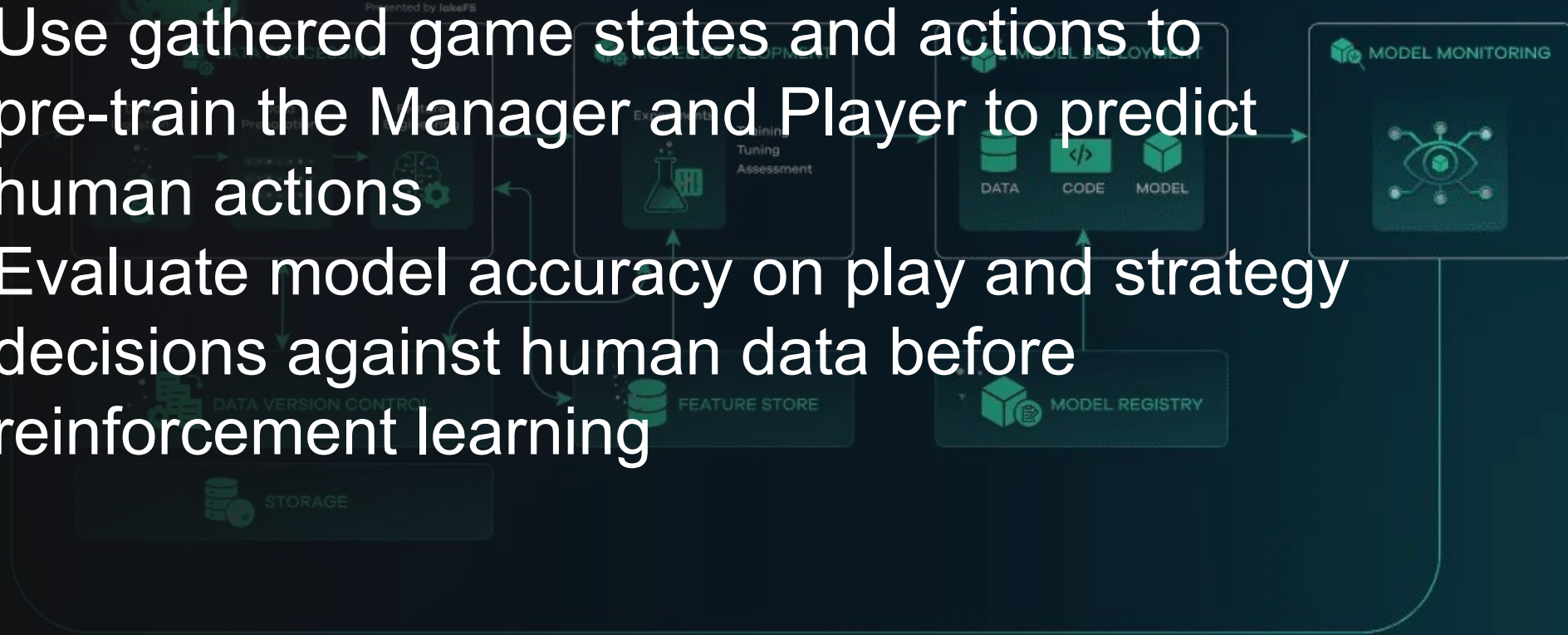
Start

x_t

Current game state info

Methodology: Model Training Part 1

- Use gathered game states and actions to pre-train the Manager and Player to predict human actions
- Evaluate model accuracy on play and strategy decisions against human data before reinforcement learning



Methodology: Model Training Part 2

- Have Player and Manager use modified API to start playing Balatro
- Rewards
 - Ante score - Player (moment to moment)
 - Blinds/Rounds Finished - Manager (Long Term)
- MVP goal:
 - Have the bot beat the second blind (6 rounds)

Actions

Go left

Go right

Jump

Stay still

Environment

Rewards (positive)

Largest reward

Maximise t

Choose the action to maximise rewards based
on state of agent

Methodology: Analysis

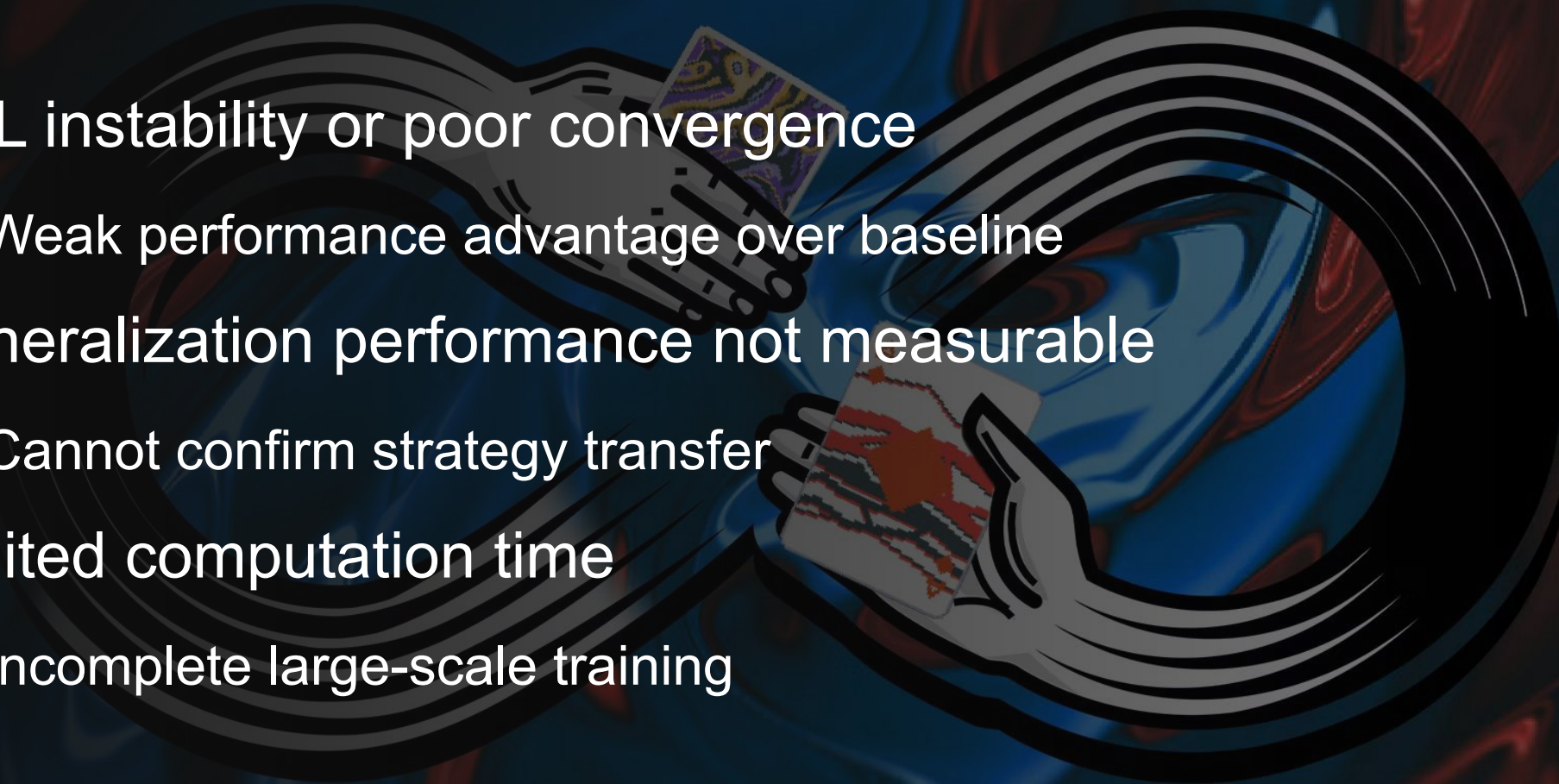
- Study logs of model games to find strategies employed by the model
 - Hands played, jokers bought, highest score, best hand played
- Compare combination of Manager and Player against just the Player and Human players
- Use reliable as a more rigorous way of performance analysis

Timeline

Task	Description	Deadline
Project Proposal	Complete overall project plan (<i>Done</i>)	Aug 17
API Data Gathering	Edit API to send all required game data to Python; re-ordering fix pending(<i>Mostly done</i>)	Aug 18
API Sending	Edit API to receive actions from Python scripts	Aug 22
Data Processing	Use Python (Pandas/Polars) to read API JSON data and reformat for model training	Aug 29
Model Training	Run experiments, tune hyperparameters, document methodology & preliminary results	Sep 22
Data Analysis	Analyse different results; conduct in-depth experiments; evaluate generalisation	Oct 3
Summary & Final Report	Interpret results, compile final report, prepare and deliver final presentation	Oct 13

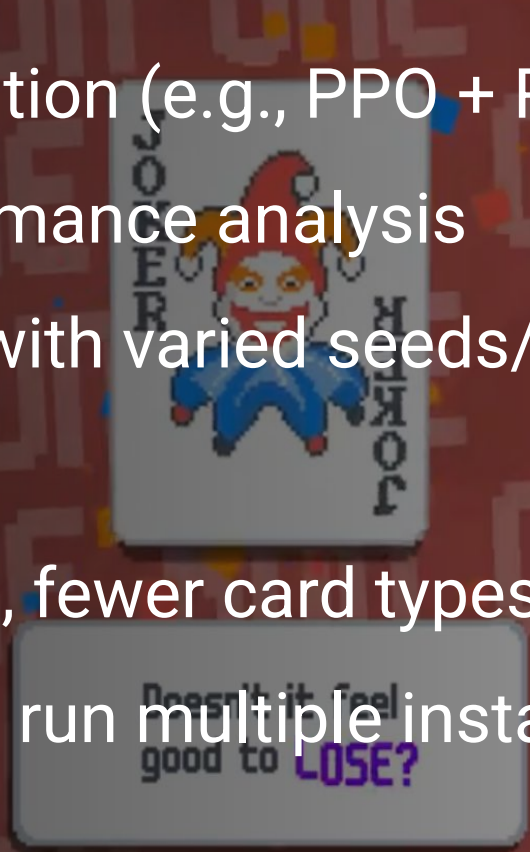
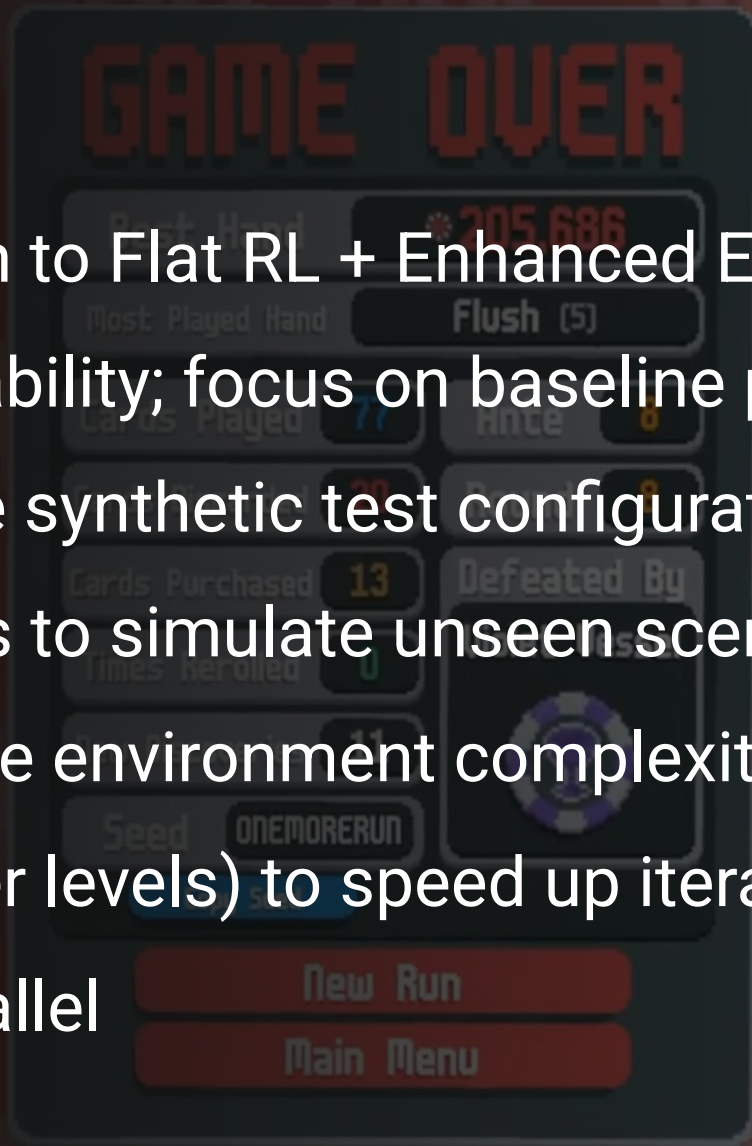
Risks

- HRL instability or poor convergence
 - Weak performance advantage over baseline
- Generalization performance not measurable
 - Cannot confirm strategy transfer
- Limited computation time
 - Incomplete large-scale training



Plan B

- Switch to Flat RL + Enhanced Exploration (e.g., PPO + RND) for stability; focus on baseline performance analysis
- Create synthetic test configurations with varied seeds/card setups to simulate unseen scenarios
- Reduce environment complexity (e.g., fewer card types, shorter levels) to speed up iterations; run multiple instances in parallel



Questions?

